1 Planning (15 marks)

Defining the problem (3 marks)

P *t* is the independent variable or vary *t*. [1]

P *V* is the dependent variable or measure *V*. [1]

P Keep the current (in the primary coil) constant. [1]

Methods of data collection (5 marks)

M Diagram showing two independent labelled coils wound on iron cores. [1]

M AC power supply / signal generator connected to one coil. [1]

M Voltmeter / oscilloscope connected to other coil in a workable circuit. [1]

M Measure thickness of card using micrometer / vernier calipers / digital calipers. [1]

M Method to keep current constant – rheostat (or variable power supply) and ammeter correctly positioned in primary circuit and explained. Diagram and text required. [1]

Method of analysis (2 marks)

M Plot a graph of ln *V* against *t* (allow lg *V* against *t*) or ln V / V_0 against *t* [1]

M σ = – gradient [1]

Safety considerations (1 mark)

S Precaution linked to hot coil(s) e.g. switch off when not in use / do not touch / wear gloves. [1]

Additional detail (4 marks)

D Relevant points might include [4]

1 Use large current (in primary coil)/large number of turns on the secondary to achieve measurable *V* (allow more turns on secondary than primary).
2 Keep frequency of power supply constant or keep the number of turns on each coil constant.
3 Use laminated cores or use insulated wire for turns.
4 Repeat measurements of *t* and average.
5 Measurement of V_0 stating that no card is present.
6 Logarithmic equation: $\ln V = \ln V_0 - \sigma t$
7 Relationship is valid if the graph is a straight line with *y*–intercept = $\ln V_0$
8 Discussion of compression of card / measure *t* when secured.

Do not allow vague computer methods.

[Total: 15]

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